Fin Econ P Set 3

1.

$$\max E\sum_{t=1}^{\infty} B^{t-1} V(C_t) \quad \text{s.t. } W_{t+1} = \left(\sum_{i=1}^{N} a_t^i R_{i,t+1} - \left(1 - \Sigma a_t^i\right) R_f\right)\cdot\left(W_t - C_t\right)$$

$$\sum_{i=1}^{N} a_t^i = 1$$

Using Bellman C_t $\quad \frac{\partial U}{\partial c_t} + B E_t\left(\frac{\partial V_{t+1}}{\partial W_{t+1}} \frac{\partial W_{t+1}}{\partial c_t}\right) = 0$

$$\frac{\partial U}{\partial c_t} = U'(C_t)$$

$$V_{t+1}(W_{t+1}) = U\left(C \sum_{i=1}^{N} a_t^i R_{t+1}^i - \left(1 - \sum_{i=1}^{N} a_t^i\right) R_f\right)\cdot W_t - C_t)$$

$$\frac{\partial V_{t+1}(W_{t+1})}{\partial V_{t+1}} = V'(W_{t+1})$$

$$\frac{\partial W_{t+1}}{\partial c_t} = -\sum_{i=1}^{N} a_t^i R_{t+1}^i - \left(1 - \Sigma a_t^i\right) R_f$$

$$V'(C_t) - B E_t\left(V'(W_{t+1}) \cdot \Sigma a_i \cdot R_{t+1}^i - \left(1 - \Sigma a_t^i\right) R_f\right) = 0$$

$E_t(R_f) = 0$

$$V'(C_t) = B E_t\left(V'(W_{t+1}) \cdot \Sigma a_t^i R_{t+1}^i\right)$$

Bellman a_t^i — $\frac{\partial V}{\partial a_{it}} + B E_t\left(\frac{\partial V}{\partial W_{t+1}} \frac{\partial W_{t+1}}{\partial a_t^i}\right) = \lambda_t$

$$B E_t\left(V'(W_{t+1}) \cdot R_{t+1}^i (W_t - C_t)\right) = \lambda_t$$

$$B(W_t - C_t) E_t\left(V'(W_{t+1}) \cdot \Sigma a_t^i R_{t+1}^i\right) = \lambda_t$$

used the fact that $a_t^i = 1$

substitute $V'(C_t)$ into equation

$$(W_t - C_t)\, V'(C_t) = \lambda_t$$

$$(W_t - C_t)\, V'(C_t) = B E_t\left(V'(W_{t+1}) \cdot R_{t+1}^i\right)(W_t - C_t)$$

$$V'(C_t) = B E_t\left(V'(W_{t+1}) \cdot R_{t+1}^i\right)$$

$$V'(W_{t+1}) = U(C_{t+1})$$

thus

$$V'(C_t) = B E_t\left(U'(C_{t+1}) \cdot R_{t+1}^i\right)$$

Problem 2

2a $\max_{a} E(U^j) = \max_{a} E(-e^{-bw_1^j}) = \max E(w_1^j)$

$w_1^j = (\alpha R_1^R + (w_0 - e_j - \alpha) R_f + R_1^V \cdot e_j$

$\max E(w_1^j) \Rightarrow E(w_1^j) - \frac{b}{2} Var(w_1^j)$

$= E(\alpha R_1^R + (w_0 - e_j - \alpha) R_f + (R_1^V e^j)$

$- \frac{B}{2} Var(\alpha R_1^R + (w_0 - e_j - \alpha) R_f + R_1^V e^j)$

$E(w_1^j) = \alpha u + (w_0 - e_j - \alpha) R_f + e_j u$

$Var(w_1^j) = \alpha^2 \sigma_R^2 + e_j^2 \sigma_V^2 + 2 \cdot \alpha e_j \rho \cdot \sigma_R \sigma_V$

$\max \alpha u + (w_0 - e_j - \alpha) R_f + e_j u - \frac{b}{2}[\alpha^2 \sigma_R^2 + e_j^2 \sigma_V^2 + 2\alpha e_j \rho \sigma_R \sigma_V$

FOC α: $M - R_f - b e_j \rho \sigma_R \sigma_V - \alpha b \sigma_R^2 = 0$

$$\alpha = \frac{-R_f + u - b e_j \rho \sigma_R \sigma_V}{b \sigma_R^2}$$

2b as ρ increases we will decrease α in the optimal allocation.

as σ_V^2 increases we also decrease optimal allocation α. Same is true of e_V.

2c.

Merton α: $\dfrac{E(r_1) + \frac{1}{2}\sigma_R^2 - R_f}{\gamma \sigma_r^2}$

The solution in 2a is very similar except we now have $-2b\rho\sigma_R\sigma_V$ in the numerator and remove $\frac{1}{2}\sigma_R^2$. We have to take asset V into account and the result is different since we use a different utility function.

2d

Solving for $\mu - R_f$ risk premium

$$\sum a_j = 1 = \sum \frac{\mu - R_f - be_j\rho\sigma_R\sigma_V}{b\sigma_R^2} = 1$$

$$= \mu \cdot J + R_f \cdot J - J \cdot b \cdot \rho \cdot \sigma_R \cdot \sigma_V \cdot \sum e_j = b\sigma_R^2$$

$$\mu - R_f = \frac{b\sigma_R^2}{J} + b \cdot \rho \cdot \sigma_R \cdot \sigma_V \cdot \sum e_j$$

$\mu - R_f$ depends on ρ and σ_V^2 when $\rho \neq 0$. So asset R is relatively more risky when ρ increases or σ_V^2 increases when $\rho \neq 0$.

Problem 3

Bellman $$V_t(W_t) = \max_{C_t, \alpha_t} U(C_t) + E_t\left(\beta V_{t+1}(W_{t+1})\right)$$

$$\tilde{W}_{t+1} = \frac{W_{t+1}}{Exp(\rho_{t+1})} = \frac{\left(\alpha_t R_{t+1} + (1-\alpha_t) R_f\right)(W_t - C_t) + y_{t+1}}{Exp(\rho_{t+1})}$$

$$y_{t+1} = e^{f_{t+1}} e^{\rho_{t+1}} e^{v_{t+1}}$$

$$\tilde{W}_{t+1} = \frac{\left(\alpha_t R_{t+1} + (1-\alpha) R_f\right)(W_t - C_t)}{exp(\rho_{t+1})} + exp(f_{(t+1)} + v_{t+1})$$

$$exp(\rho_{t+1}) = exp(\rho_t)\, exp(z_{t+1})$$

$$\tilde{W}_{t+1} = \frac{\alpha_t R_{t+1} + (1-\alpha) R_f (\tilde{W}_t - \tilde{C}_t)}{exp(z_{t+1})} + exp(f(t+1) + v_{t+1}$$

$$U(\tilde{C}_t) = U(C_t) \cdot exp(\rho_t)^{1-\gamma}$$

$$V_t(\tilde{W}_t) = \max_{\tilde{C}_t, \alpha_t} \frac{\tilde{C}^{1-\gamma}}{1-\gamma} \cdot exp(\rho_t)^{1-\gamma} + \beta E\ V_{t+1}(\tilde{W}_{t+1})$$

$$V_{t+1}(W_{t+1}) = V_{t+1}\left(\tilde{W}_{t+1} \cdot exp(\rho_{t+1})\right) = exp(\rho_{t+1})^{1-\gamma} \cdot V_{t+1}(\tilde{W}_{t+1})$$

$$exp(\rho_{t+1}) = exp(\rho_t) \cdot exp(z_{t+1})$$

$$V_{t+1}(\tilde{w}_{t+1}) \cdot (\exp(\rho_{t+1}))^{1-\gamma} = V_{t+1}(\tilde{w}_{t+1}) \cdot (\exp(\rho_t) \cdot \exp(z_{t+1}))^{1-\gamma}$$

$$V_t(\tilde{w}_t) = \max_{\tilde{c}_t, \alpha_t} \frac{\tilde{c}^{1-\gamma}}{1-\gamma} \cdot \exp(\rho_t)^{1-\gamma} + \beta E\left(\exp(\rho_t)^{1-\gamma} \exp(z_{t+1})^{1-\gamma} V_{t+1}(\tilde{w}_{t+1})\right)$$

we can divide by $\exp(\rho_t)^{1-\gamma}$ since it doesn't change $\tilde{c}_t^*, \alpha_t^*$

$$V_t(\tilde{w}_t) = \max_{\tilde{c}_t, \alpha_t} \left(\frac{\tilde{c}^{1-\gamma}}{1-\gamma} + \beta E\left(\exp(z_{t+1})^{1-\gamma} V_{t+1}(\tilde{w}_{t+1})\right) \right)$$